

TrES-3b

R-0120

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240306 · created 2026-07-16T05:01:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460375.9249 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.85 +/- 2.25 [%]
Orbital Inclination (inc)	86.89 +/- 2.89
Airmass coefficient 1 (a1)	1.53 +/- 0.48
Airmass coefficient 2 (a2)	-0.359 +/- 0.086
Scatter in the residuals of the lightcurve fit is	24.65 %
Best Comparison Star	#1 - [618, 454]
Optimal Aperture	3.98
Optimal Annulus	4.68
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.057 vs 0.16309 (deviation 0.47 σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 1.73 σ)
Mid-time O-C	-5.2 min
Depth SNR	2.8
Residual scatter	24.65 %

Light curve

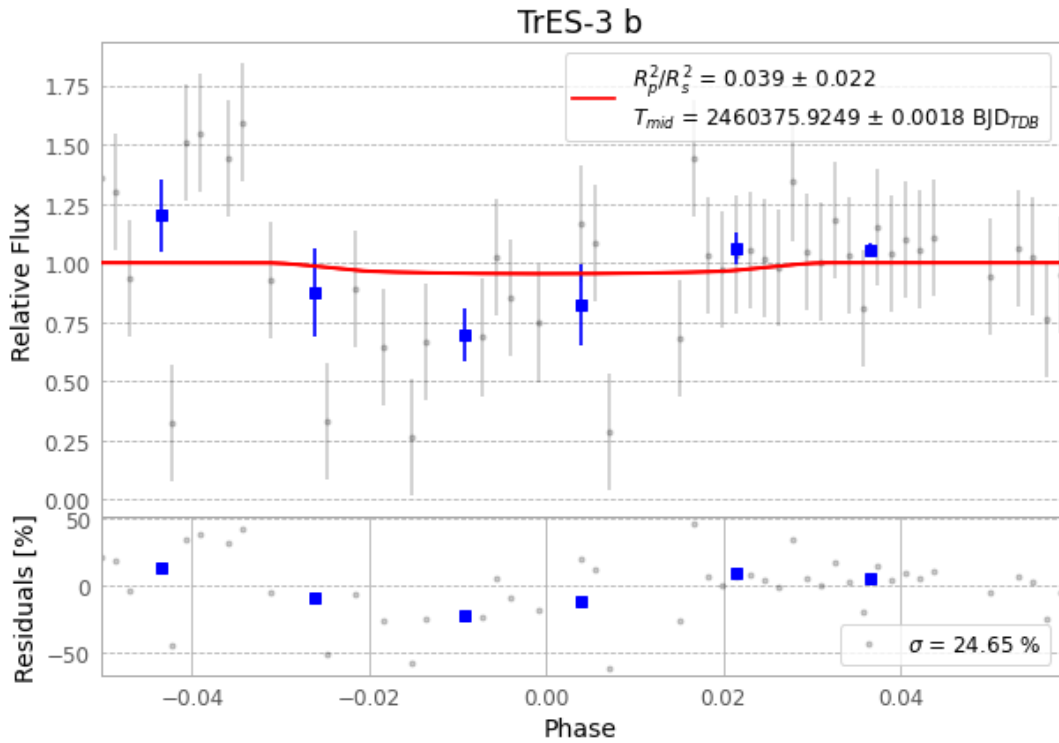


Figure 1 — FinalLightCurve_TrES-3 b_2024-03-06.png (SHA-256 c81a265a4e374d72...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f1d424c56cf9d7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

69 FITS frames (45 MB), TRES-3240306083615.FITS → TRES-3240306120015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240306.log (8b42d96c7b5b...), FinalLightCurve_TrES-3 b_2024-03-06.png (c81a265a4e37...), FinalLightCurve_TrES-3 b_2024-03-06.csv (9802c92923fe...)

Compute resources

Wall time 190.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:01Z.